# Compositional Processes as Graph

## Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

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**Question.** How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

**A possible answer (from the AWG case).** Relations that do not force one sequence, uncertainty marked on the statement rather than in the relation, and generated and editorially asserted knowledge kept apart. They help in part: what a model has no form for stays in prose – readable, but not queryable.

## 1 The AWG: three forms of edition, one graph in two layers

| Form of edition | Presents | Edits |
|---|---|---|
| work edition · Werkedition | versions published or meant for print | `TextVersion` |
| text edition · Textedition | versions beyond a sketch, not printed | `TextVersion` |
| sketch edition · Skizzenedition | diplomatic transcriptions of sketches | `Sketch` |

**Generated layer**
derived from the edition data

**Curated layer**
asserted by the editors

Draft ontology v0.1 · SPARQL in the app · Turtle export · no reasoning

## 2 Op. 25: the sketches cross the opus order

Op. 25 = M 317 (song I) · M 322 (song II) · M 321 (song III). The curated chain:

**M 317 Sk4** → **M 321 Sk1–9** → **M 322 Sk1**

Song III was sketched before song II. The generated layer shows three separate “islands”.

## 3 Terms from the material become classes

| Term | In the documents | Class |
|---|---|---|
| sketch · Skizze | Sk1, Sk2, Sk2.1 | `Sketch` ⊂ `Inscription` |
| row table · Reihentabelle | Webern’s sheet of row forms | `RowTable` ⊂ `Sketch` |
| paratext · Paratext | title, dating, instrumentation | `Paratext` ⊂ `Inscription` |
| version · Textfassung | TF1, TF2 | `TextVersion` |
| edition complex · Editionskomplex | all edition-relevant material of a composition, by opus or M number | `EditionComplex` |
| uncertainty | “presumably” | tag `certainty: uncertain` |

Not modelled yet: “study composition”, “ink draft”.

## 4 Two relations link inscriptions: a sequence, or mutual influence

| Relation | Meaning | cf. CIDOC CRM |
|---|---|---|
| `precedes` | A begun before B, overlap allowed | P176 “starts before the start of” |
| `concomitates` | A and B shape each other; which came first cannot be decided | P174 “starts before the end of” (the overlap, not the influence) |
| `precedesScripture` | evidence: the writing | |
| `precedesContent` | evidence: the musical content | |

`precedes` is transitive; `concomitates` is symmetric, not transitive.

## 5 For now, the model is domain-specific, with links to standards

**Relating sketches draws on content and page together** – which LRMoo separates (F2 Expression vs. F5 Item).

| Standard | In the AWG model | Link |
|---|---|---|
| LRMoo | edition complex ≈ F1 · version, sketch ≈ F2 · document ≈ F5 | `rdfs:seeAlso` |
| CIDOC CRM | temporal reading: `precedes` ≈ P176 · `concomitates` ≈ P174 | `rdfs:seeAlso` |
| DC terms, SKOS, PROV, `rdf:Statement` | single terms: parts · concept schemes · attribution · statements | used directly |
| MEI | not integrated yet | – |

## 6 The model could transfer; the pipeline cannot

**Portable**

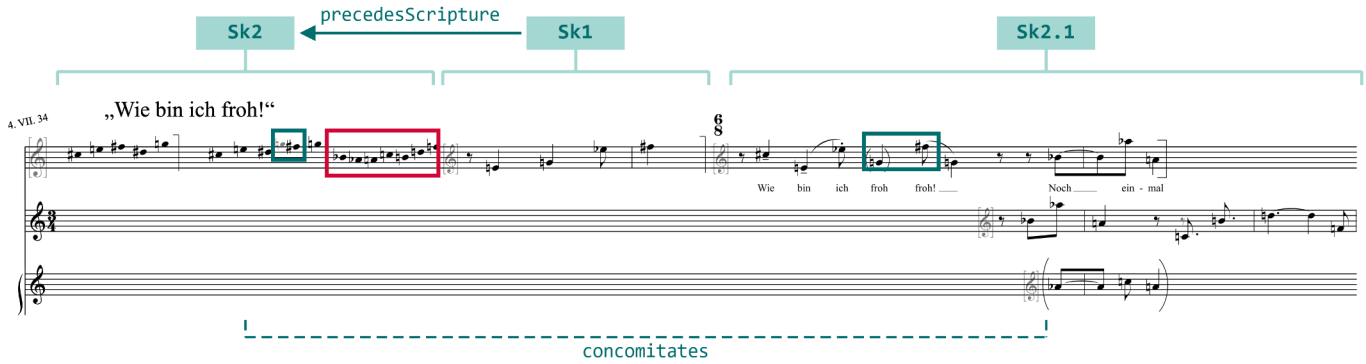
- `Inscription` · `Document` · `Folio` · `Placement`
- `precedes` · `concomitates`

**Tied to the AWG**

- the generator reads the AWG's own data and M numbers
- Sk1 before Sk2 is read as `precedes`, without evidence type

Transfer to other repertoires is plausible but untested.

## 7 Example: M 317, sketchbook 3, fol. 38v




**Sk1 `precedesScripture` Sk2**
The end of Sk2 is squeezed (red): Sk1 already took the space – an order inferred from the writing.

**Sk2.1 `concomitates` Sk2**
In both, g gives way to f♯ (green): which came first cannot be decided.

Source: AWG online edition, M 317 Sk2, Sk1, Sk2.1, superimposed (development version, 12 Nov 2026).

## 8 What the graph shows

**→ Clusters**
Sk2.1 and its variants: one work phase at a glance

**→ Paths**
Sk1 → Sk4 → M 321: the work leaves the song

**→ Gaps**
a “presumably” without a tag counts as certain

## 9 Where model and data diverge, each gap is information

**Model lags data**

- staves, measures in their own notation, stored as strings
- no writing instruments yet

**Data lags model**

- sketchbook 3 as two excerpts · 29 Sk## placeholders
- datings without year

Some gaps are ours to close; others belong to the evidence. The generator reports them: 12 warning rules.

## 10 Conclusion

- **Layers** show where editorial judgment becomes explicit.
- **Warnings** show where the data are open or disagree.
- **Model and data** are written side by side: concomitant.

Next: named graphs for ink and pencil layers · persons · archives

**For discussion.** Whose process does the graph show – Webern's or ours?
How much of (Webern's) compositional processes survives translation into formal models?